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Effects of Sedimentation and Lesion Position on Coral Tissue Regeneration

E. H. MEESTERS, A. BOS, AND G. J. GAST

Carmabi Institute, PO Box 2090, Curaçao, Netherlands Antilles
Institute of Taxonomic Zoology, University of Amsterdam, PO Box 4766, 1009 AT Amsterdam, The Netherlands

Abstract. The tissue regeneration potential of various coral species was studied on the leeward fringing reefs of Curaçao, Netherlands Antilles. Regeneration of *Acropora palmata*, a branching species, common in shallow parts of the reef, was slower in colonies subjected to long-term increased levels of sedimentation than in control colonies. *Diploria strigosa*, *Porites astreoides*, and *Siderastrea siderea*, were not as sensitive as *A. palmata*. Regeneration of *Montastrea annularis*, the main reef-building coral in the Caribbean, was slower for colonies that suffered from high levels of sedimentation, caused by the construction of an artificial beach. Whether a lesion was positioned at the top or at the side of a colony, had no influence on lesion regeneration, even though side lesions received approximately 70% less light. Photoadaptation within a colony was considered to be responsible for the lack of a regeneration difference between lesion positions, as chlorophyll a concentrations in polyps on the sides of colonies were twice as high than in polyps on the top.

Introduction

Corals are frequently damaged by biotic and abiotic causes (reviewed by Pearson 1981). Large areas are devastated by catastrophes, and reported as being devoid of living coral cover. However, when small areas are examined closely, it is often found that parts of colonies have survived (Connell 1973, Highsmith et al. 1980, Knowlton et al. 1981). Partial mortality, a characteristic phenomenon in clonal organisms, will often result in a small patch of bare skeleton (partly) surrounded by living tissue. These patches may become colonized by algae, boring sponges and other organisms, but often they will regenerate from the surrounding tissue (Bak et al. 1977, Bak and Steward-van Es 1980, Fishelson 1973).

Coral species regenerate lesions at different rates, and this has been related to life history strategies (Bak 1983). The rate at which a lesion recovers also appears to decrease exponentially through time (Bak 1983). Because the energy surplus of corals appears to be limited (Davies 1991, Muscatine et al. 1984), we hypothesized that stress, reducing the amount of available energy, will also inhibit the regeneration of small lesions.

Increased sedimentation is often cited as an important cause of coral death or changes in community structure (Pearson 1981). Its effect was studied *in situ* on different coral species in two experiments: one, with long-term sedimentation stress, and another with increased sedimentation for only a short time.

Apart from species specific characteristics that influence the regeneration of small lesions, regeneration may also be affected by the position of the damage. Since light intensities vary substantially over the surface of a single colony (Roos 1967), we hypothesised that lesions on the side of a colony will regenerate slower than lesions on top of a colony, because colony sides will receives less light than colony tops.

Materials and Methods

Study Sites

The four study sites were situated on the leeward fringing reefs of Curaçao, Netherlands Antilles (Fig.

1). All are characterized by a submarine terrace normally stretching a distance of 50 to 100 m from the coast to a depth of 7 to 12 m, where the reef steeply slopes to a depth of 50 to 60 m (Bak 1975). Abundant corals on the shallow terrace (Bak 1977) are *Acropora palmata* (Lamarck), *Agaricia* spp., *Diploria clivosa* (Ellis and Solander), *D. strigosa* (Dana), *Montastrea annularis* (Ellis and Solander), *M. cavernosa* (Linnaeus), *Porites astreoides* Lamarck, and *Meandrina meandrites* (Linnaeus).

Lesion Infliction

Lesions were made on top of haphazardly selected colonies with a submersible drill powered by compressed air and equipped with a small grinding point that created flat circular patches of scraped skeleton, devoid of living tissue, of approximately 1 cm diameter and 2 mm depth. In all experiments, lesion area was measured with a plexiglass ruler divided into 0.04 cm^2 squares. At the start of the experiments lesion depth was measured with a pair of callipers to calculate the surface area of the lesion wall.

Long-term Sedimentation

To study the effect of prolonged sediment stress on the regeneration rate of corals, lesions were made on colonies of *Acropora palmata*, *Diploria strigosa*, *Porites astreoides* and *Siderastrea siderea* (Ellis and Solander). The two study sites, Low and High, are similar in important environmental characteristics, such as nutrient concentrations, wave exposure and current, but different based on sedimentation data and light extinction measurements (Fig. 1). They have an average sedimentation rate of 27, and 40 g dry weight m^{-2} d^{-1}, respectively ($P = .0015$, comparison of monthly values with Wilcoxon's signed-ranks test). Sedimentation data were collected using sediment traps with length to width ratio of 10. The traps were placed on the shallow terrace (5 m) with openings approximately 1 m above reef bottom, collected on a monthly basis during a period of one year, filtered over Whatman GFC filters and dried to constant weight at 100°C. Incidental light extinction measurements were taken with an International Light photometer, IL 1400A. Extinction coefficients for site High were sometimes twice as high as for site Low, ranging respectively from −0.12 to −0.26, and −0.10 to −0.13 (Means ± SE: −0.179 ± 0.045 and −0.118 ± 0.002).

Lesions on branches of *Acropora palmata* colonies were made on the upper side at a distance of 10–20 cm from the edge (7/12/90; $N_{Low} = 30$, $N_{High} = 35$; depth = 3–4 m). Colonies of *Diploria strigosa* received a lesion in between two ridges (9/4/90; $N_{Low} = 30$, $N_{High} = 29$; depth = 4–6 m). *Porites astreoides* lesions were made on top of every colony (12/5/90; $N_{Low} = 30$, $N_{High} = 34$; depth = 4–6 m). Lesions on *Siderastrea siderea* were made on the top of the colony (6/7/91, $N_{Low} = 28$; 6/10/91, $N_{High} = 29$; depth = 4–6 m).

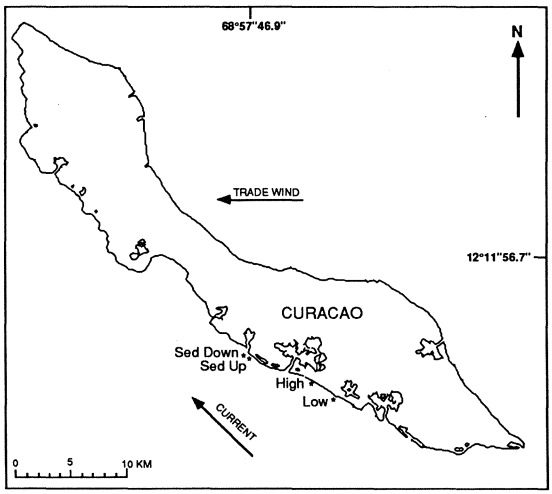

Fig. 1. The island of Curaçao and situation of the study sites. Site Low and High are the respectively low and high sedimentation sites of the long-term stress experiment. SedDown and SedUp are the sites of the short-term sediment stress experiment, down-current and up-current of the artificial beach construction site. Main wind and current direction are shown.

Short-Term Sedimentation

To assess the influence of sudden sediment stress on the regeneration rate of corals, lesions were inflicted on *Montastrea annularis* colonies following the start of the construction of an artificial beach in October 1991. The experimental colonies were situated 100 m upcurrent and 50–75 m downcurrent of the construction site (Fig. 1; SedDown, and SedUp, 10/16/91; $N_{SedDown} = 23$, $N_{SedUp} = 30$). Sedimentation at each site was measured by placing two sediment traps (length/ width ratio = 10) between the experimental colonies. Traps were collected at a 2 weekly interval and the contents was processed as explained in the previous section.

Lesion position

To assess the influence of lesion position on regeneration, lesions were inflicted on the tops and sides

of separate colonies of *P. astreoides* (Site Low, N_{top} = 24, N_{side} = 22). Light measurements were taken near every lesion with an International Light photometer, IL 1400A. Samples were taken at the end of the experiment to assess chlorophyll a concentration. Because regenerated lesions remained lightly colored during a long time (indicating changed pigment concentrations) samples were taken directly next to the lesions. Sample processing and chlorophyll measurements were as described by Szmant and Gassman (1990).

Analysis

Lesion size data (mm^2) were logarithmically transformed (X+1) and regressed against time. Not all measuring dates could be used in the regression analysis, because the lesion sizes were no longer normally distributed towards the end of the experiments, but skewed to the right, as the number of closed lesions increased. Testing for significance of regression was done by calculating the variance ratio of the variation due to linear regression over the variation due to deviations from regression. This reduces the influence of heterogeneous data and results in a more conservative test (Sokal and Rohlf 1981). Because regeneration rate is not constant, but decreases with time, it is best described by the slope of the exponential regression line. The different treatments were compared by testing for difference between regression slopes.

Results

Long-Term Sedimentation

Regeneration rate clearly depended on the species involved. *A. palmata* regenerated more slowly at elevated sedimentation levels, while regeneration rates of *Porites astreoides* and *Siderastrea siderea* were not different at the two sites (Fig. 2, Table 1). The regeneration rate of *Diploria strigosa* colonies was faster under conditions of high sedimentation (Fig. 2, Table 1). The regression however explains only a very small part of the variation (R^2 in Table 1). This is mostly caused by 8 lesions at site Low that did not regenerate, but remained approximately the same size. Two of these were excluded from the regression calculation of site Low (Table 1) because they enlarged to fourfold their size as a result of fish grazing. Removing of the other 6 lesions resulted in a non-significant difference between the two regression slopes. Possibly, these were also the result of fish activities.

A. palmata quickly covered the bare skeleton with a thin veneer of transparent undifferentiated tissue. Often the lesion was already closed before any polyps were formed in the new tissue. Polyp formation started after approximately 8 days, and was visible in almost all lesions after 18 days, probably associated with the developmental phase of the newly formed tissue. Corallite development quickly followed the appearance of a polyp mouth. New corallites were initially differently oriented from the surrounding corallites. After the lesion had fully regenerated, the tissue remained slightly depressed and lighter colored than the surrounding tissue. Two months after closing, nothing remained that could be attributed to a lesion.

Regeneration in *P. astreoides* was also very fast. Again a thin layer of tissue grew inwardly, although more slowly than in *A. palmata*. Polyps formed almost immediately in the new tissue. Some lesions were also closed from within the lesion, suggesting that remaining tissue in the perforate skeleton may accelerate the regeneration of shallow lesions. The depression that remained after the lesion had been closed disappeared in the next months.

In *Diploria strigosa* lesion regeneration rate was slower than in the previous species. At first the lesion was covered by undifferentiated tissue, which was already colored, although lighter than the surrounding tissue. The position of many healed lesions remained visible because the ridges surrounding the lesion tend to form a star-shaped pattern on the colony surface.

Regeneration was extremely slow for *Siderastrea siderea*. Tissue grew inwardly from the edge of the lesion, and polyps were formed simultaneously. Polyps on the edges of the lesions, were also observed to elongate septa towards the middle of the lesion. After 128 days 75% and 85% of all lesions were still open on respectively site Low and site High.

Short-Term Sedimentation

Average sedimentation (g dry weight m^{-2} d^{-1}; N = 2) during the the first, second and third 14 days was 24 vs 48 (P = .009, one-tailed t-test), 31 vs 37 (P = .049), and 83 vs 107 (P = .088) respectively for upcurrent vs downcurrent of the construction site. Extremely high sedimentation in the last period was due to the building of a new pier, which was very close to the downcurrent site. Sedimentation at the downcurrent site was very variable, due to the varying intensity of construction activities. It was also noted that suspended sediment tended to stay close to the shore for a long distance, due to the direction of incoming waves. Sedimentation at 50 m from

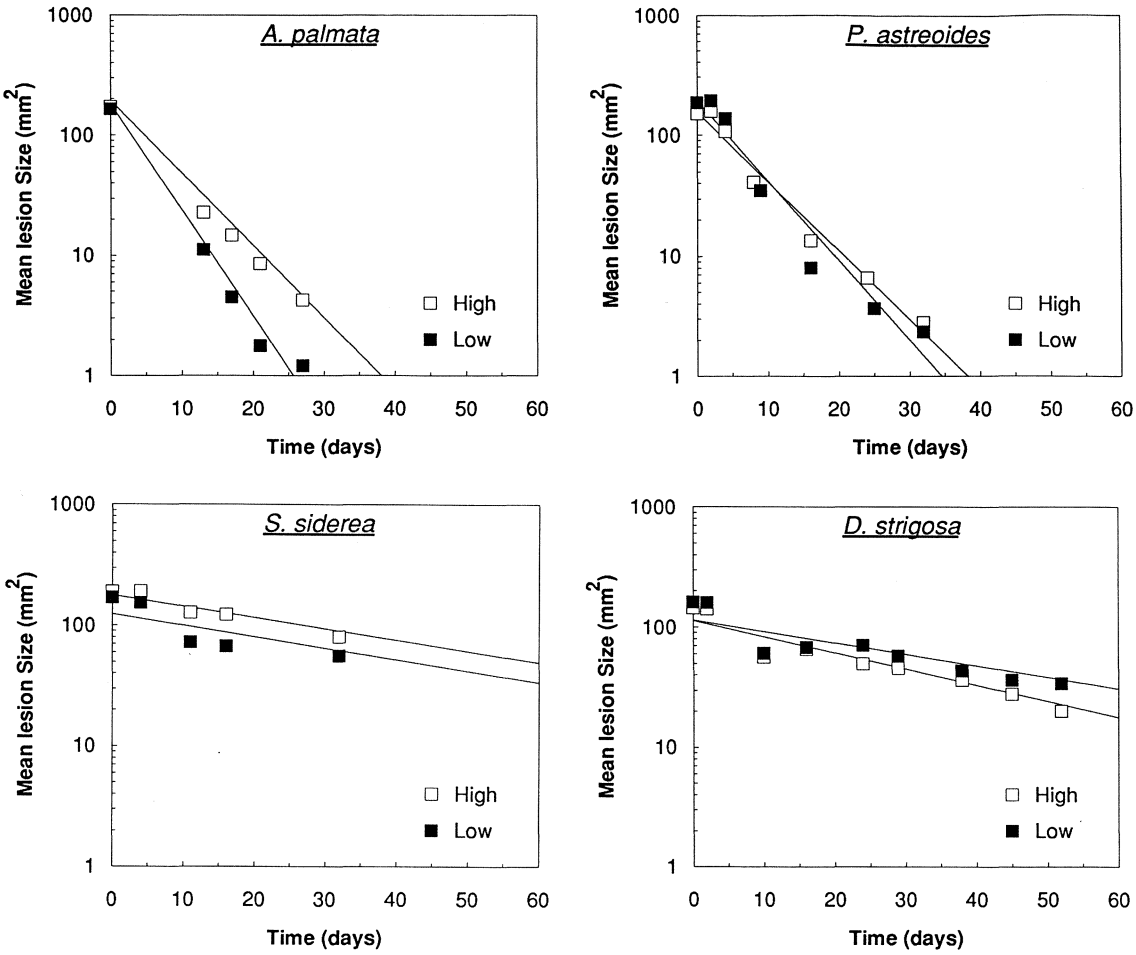

Fig. 2. Regeneration of the four species used in the long-term sediment stress experiment. Lesion size is given as the mean size of all lesions at a certain time. Fitted regression lines are drawn in graphs. Low: low sedimentation site; High: high sedimentation site.

the shore was 25% higher than at 100 m (Meesters unpub.).

M. annularis regenerated more slowly at the high sedimentation site (SedDown) than at the low sedimentation site (SedUp) (Fig. 3, Table 1). Regeneration of *Montastrea annularis* lesions was characterized by elongating septa of partly damaged polyps and the inward growth of undifferentiated tissue. No mortality of corals was observed up to two months after the start of construction activities. At that time many species started to show signs of partial discoloration.

Lesion Position

Lesion position in *P. astreoides* did not influence regeneration rate (Fig. 5, Table 1). Light measurements showed that the lesions on the side of the colony received on the average 30% of the amount of light that reaches the top of the colony (Mean ± SE: 244 ± 50 Eμ m^{-2} s^{-1} and 829 ± 24 Eμ m^{-2} s^{-1}, respectively; $P = 0.0004$, MU-test). Irradiance was also influenced by surrounding objects and sand reflection.

Average chlorophyll *a* concentration of top lesions was significantly lower when compared with side lesions (respectively 4.5 ± 0.8 μg cm^{-2} and 9.3 ± 1.1 μg cm^{-2} (mean ± SE); $P = 0.004$, MU-test).

Discussion

Regeneration in *Acropora palmata* was slower at the site with a high sediment load.

A. palmata has been reported as a very inefficient sediment rejector both in the laboratory (Hubbard and Pocock 1972, Bak and Elgershuizen 1976) and

Table 1. Results of regression analyses and slope comparisons. Data were fitted to the line: Size = C * $10^{(b*Time)}$, in which C = constant, and b = slope. X_0 = number of days till lesion closure according to regression line, N = number of data, R^2 = coefficient of determination; Days = number of days included in regression; $P_{regr.}$ = Probability of regression, $P_{comp.}$ = Probability of no difference between slopes: *** = $P < 0.001$, ** = $P < 0.005$, * = $P < 0.025$, n.s. = not significant.

Species	Site	C	Slope	X_0	N	R^2	Days	$P_{regr.}$	$P_{comp.}$
Long-term sedimentation									
Acropora palmata	Low	182	−0.088	26	164	0.86	27	***	*
	High	192	−0.060	38	244	0.65	36	***	
Diploria strigosa	Low	114	−0.010	206	308	0.22	98	***	**
	High	112	−0.013	158	319	0.47	98	***	
Porites astreoides	Low	191	−0.066	35	186	0.75	32	***	n.s.
	High	154	−0.057	38	237	0.74	32	***	
Siderastrea siderea	Low	124	−0.0095	220	196	0.61	128	***	n.s.
	High	178	−0.0094	239	203	0.65	128	***	
Short-term sedimentation									
Montastrea annularis	SedDown	162	−0.020	110	178	0.57	35	***	***
	SedUp	177	−0.030	75	260	0.75	48	***	
Lesion position									
P. astreoides Top	Low	205	−0.044	53	191	0.74	34	***	n.s.
Side	Low	202	−0.046	50	175	0.68	34	***	

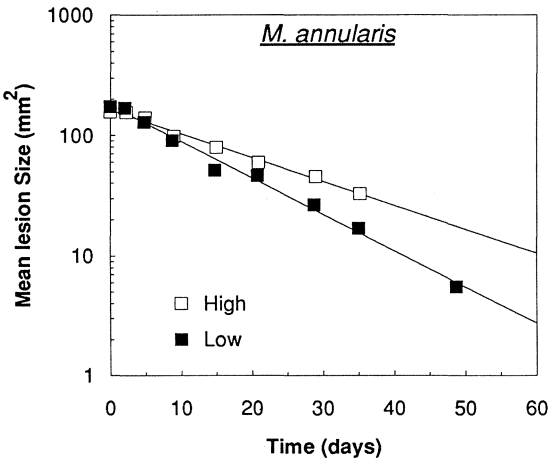

Fig. 3. Regeneration of *Montastrea annularis* up-current (SedUp) and down-current (SedDown) of the construction site of an articial beach. Lesion size is the mean size of all lesions at a given time. Fitted regression lines are drawn in graph.

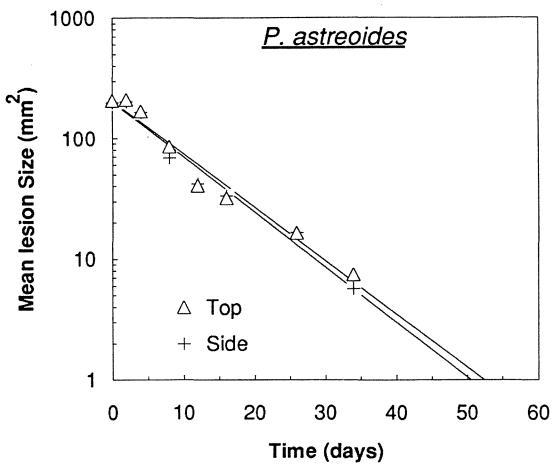

Fig. 4. Regeneration of lesions on the side and at the top of *Porites astreoides* colonies. Lesion size is given as the mean size of all lesions. Fitted regression lines are shown in graph.

in the field (Rogers 1983). The absence of *A. palmata* in surveys of reefs with a high sediment load further supports the notion of the relative sensitivity of this species (Tomascik and Sander 1987, pers. obs.). Sedimentation data for the high sedimentation (site High) were low compared with the results of Tomascik and Sander (1987), even though their sediment traps were of comparable size and situated at the same height above the bottom. Adey et al. (1977) attributed the disappearance of a shallow-water *A. palmata* barrier reef community to increased water turbidity which was caused by the erosion of the soil cover on a partly exposed shelf.

Regeneration in the other experimental species, *Diploria strigosa*, *Siderastrea siderea* and *Porites astreoides*, was not slower at the high sedimentation site than at the low sedimentation site. This agrees with other field observations (Loya 1976, Morelock et al. 1983) in which these 3 species, together with *Montastrea cavernosa* and *Meandrina meandrites* were relatively abundant in prolonged sediment stress situations. Such spatial dominance was also apparent on the high sedimentation reef in Curaçao (Meesters unpub.).

The regeneration rate of *Montastrea annularis* was lower at the site exposed to a short-term in-

Table 2. Life history characteristics. Comparison of regeneration, reproduction, growth, polyp position, sediment rejection, and competitive characteristics of the species used in this study. Regeneration, sediment rejection, and competitive ability are given in relative units (larger circles are faster or stronger). Reproduction: H. Bc.—hermaphroditic broadcaster; G. Bc.—gonochoric broadcaster; H. B.—hermaphroditic brooder. Polyp position shows positioning of polyp in skeleton.

	Reg.	Repr.	Growth (mm y^{-1})[a]	Polyp position	Sediment rejection[d]	Comp.[e]
A. palmata	●	H. Bc.	88.0[b]		•	●
S. siderea	•	G. Bc.	4.8		●	•
P. astreoides	●	H. B.	5.3		•	•
D. strigosa	•	H. Bc.	5.3		●	●
M. annularis	●	H. Bc.	15.5[c]		●	●

[a]Guzmán et al. 1991.
[b]Bak 1983.
[c]Tomascik 1990, columnar morphotype.
[d]Bak and Elgershuizen 1976.
[e]Lang 1973.

crease in sedimentation than at the relatively unaffected upcurrent site. This difference was apparent despite the fact that sedimentation was rather low e.g. compared to data of Tomascik and Sander (1985) for Barbados. *M. annularis* has been known to be an inefficient sediment rejector in the laboratory experiments of Bak and Elgershuizen (1976), and is apparently sensitive to small differences in sedimentation rate during a short period of time.

Increased energy expenditure on cleaning mechanisms and a reduction in photosynthetically fixed energy due to reduced light levels may cause a drop in regeneration rate under conditions of increased sedimentation. Negative effects of increased sediment load on corals have also been reported for growth rates (Dodge et al. 1974, Bak 1978, Kendall et al. 1983) and reproduction (Kojis and Quinn 1984). Bak (1983) showed that artificial lesions inhibited the growth of *A. palmata* branches, and Rinkevich and Loya (1989) found reduced fecundity in regenerating branches of *Stylophora pistillata*. This indicates that regeneration is a process at least equally important as growth and reproduction.

A lower regenerative potential may increase partial mortality, settling of other organisms on the exposed skeleton, and competition. Differential colony mortality and changes in community structure may eventually result in relative dominance by *S. siderea*, *D. strigosa*, and *P. astreoides* in terms of cover and density.

Prolonged sediment stress has a very different impact on different coral species. Short-term laboratory experiments on sediment load tolerance of corals (Hubbard and Pocock 1972, Bak and Elgershuizen 1976) provide useful information on species specific rejection capabilities, but the actual field situation can sometimes differ, because differences in wave exposure or depth influence the actual sediment load that species experience (Morelock et al. 1983, Cortes and Risk 1985).

We show that there is intra-colony photoadaptation, associated with differences in irradiance over the surface of the colony. This possibly explains the equal regeneration rates of lesions on top and side of Porites astreoides colonies. Up to now, photoadaptation in corals has mainly been studied in relation with depth (e.g. Porter et al. 1984), but it appears to be equally important within a single colony.

Of basic importance in the ecology of corals is that species differ in their ability to regenerate small lesions (Table 2). Regeneration potential of *Siderastrea siderea* is extremely low and in the field it is always the most heavily damaged species (Meesters unpub.). Damage may be caused by interspecific interactions, because *S. siderea* ranks low in Lang's "aggressive" ranking hierarchy (Lang 1973). Partial mortality in *S. siderea* may constitute a way of asexual colony propagation, especially since sexual recruitment appears to be low (Bak and Engel 1979). Regeneration in *Diploria strigosa* is also very low, but damage may be restricted mainly to the septa walls and not to the polyp mouths. Septa walls have been observed to heal faster (Meesters pers. obs.). In Porites astreoides tissue regeneration is very fast. Its low sediment rejection capacities are probably compensated for by the formation of mucus mantles (Bak and Elgershuizen 1976) that trap sedi-

ment, and are removed by water action. *P. astreoides*, *S. siderea*, and *D. strigosa* are often spatially dominant under high sediment load conditions (Morelock et al. 1983, Cortes and Risk 1985). Regeneration in *Acropora* palmata is very fast, just as its growth rate. Regeneration may benefit from the amount of energy that is spent on growth, as distance from lesion to top of branch was small.

Overall, *Montastrea annularis* emerges as best species, closely followed by *A. palmata*, *M. annularis* is the main reef-building species in the Caribbean. *A. palmata*, although much more restricted in its depth distribution, can also be a main component in reef frameworks (Adey et al. 1977).

Conclusions

1. Different species regenerate small damage at different rates.
2. Regeneration rate decreases exponentially with time.
3. Stress, such as increased sedimentation, can slow down the regeneration of damage.
4. Lesion regeneration was not influenced by differences in lesion position on the colony surface or irradiance because of intra-colony photoadaptation.

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